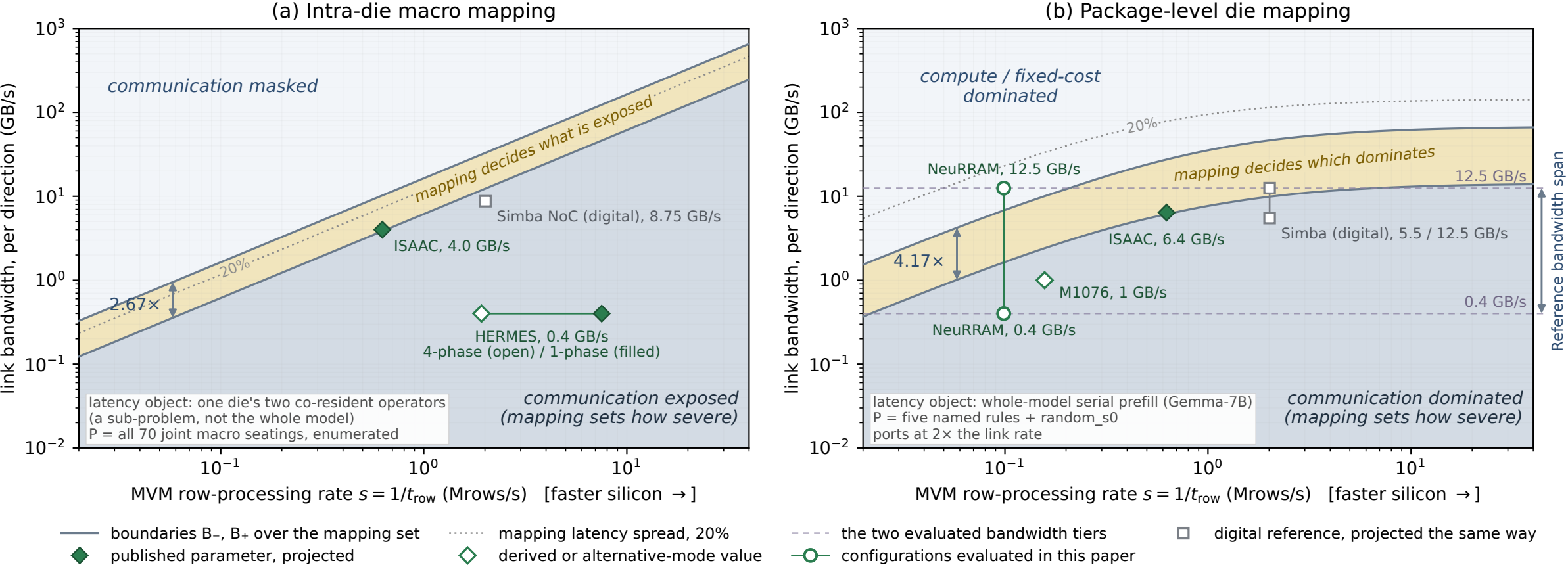
(a) Intra-die macro mapping
communication masked
mapping decides what is exposed
20%
2.67×
Simba NoC (digital), 8.75 GB/s
ISAAC, 4.0 GB/s
HERMES, 0.4 GB/s
4-phase (open) / 1-phase (filled)
latency object: one die's two co-resident operators
(a sub-problem, not the whole model)
P = all 70 joint macro seatings, enumerated
communication exposed
(mapping sets how severe)
link bandwidth, per direction (GB/s)
MVM row-processing rate $s = 1/t_{row}$ (Mrows/s) [faster silicon →]
(b) Package-level die mapping
compute / fixed-cost
dominated
20%
NeuRRAM, 12.5 GB/s
mapping decides which dominates
12.5 GB/s
Simba (digital), 5.5 / 12.5 GB/s
ISAAC, 6.4 GB/s
4.17×
M1076, 1 GB/s
0.4 GB/s
NeuRRAM, 0.4 GB/s
Reference bandwidth span
latency object: whole-model serial prefill (Gemma-7B)
P = five named rules + random_s0
ports at 2× the link rate
communication dominated
(mapping sets how severe)
link bandwidth, per direction (GB/s)
MVM row-processing rate $s = 1/t_{row}$ (Mrows/s) [faster silicon →]
boundaries B–, B+ over the mapping set
published parameter, projected
mapping latency spread, 20%
derived or alternative-mode value
the two evaluated bandwidth tiers
configurations evaluated in this paper
digital reference, projected the same way